The Art of Uncertain Topoplots: Designing and Evaluating Uncertainty Visualizations for EEG/ERP Data

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Abstract—TODO: Supplementary-material overview

Index terms—TODO: Supplementary Material

1 Supplementary Material

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ID	Interview question
Warm-up	
Q1	Could you briefly describe your background in EEG/ERP research?
Q2	What types of EEG visualizations do you most commonly use in your work, and for what purposes?
Thinking About Uncertainty	
Q3	When you think about quantifying uncertainty in EEG or ERP data, what kinds of methods come to mind?
Q4	Do you currently include uncertainty information when you visualize EEG/ERP data? If so, what type of uncertainty do you include, and why?
Use of Topoplots	
Q5	At what stages of your research do you use topoplots, and for which audiences?
Q6	When you create a topoplot, what purposes does it serve?
Q7	Why do you think other researchers use topoplots?
Uncertainty in Topoplots	
Q8	Do you ever show uncertainty in topoplots specifically? If not, why not?
Q9	If you wanted to show uncertainty in a topoplot, how might you visualize it?
Value and Barriers	
Q10	Do you think visualizing uncertainty in topoplots would be helpful? Why or why not?
Q11	Why do you think many researchers currently do not include uncertainty in their topoplots?
Future Needs	
Q12	Is there anything you wish topoplot tools did better?
Q13	What requirements would an ideal topoplot need to meet?
Closing	
Q14	Is there anything I did not ask about that you think is important regarding topoplots or uncertainty visualization?

Table S1: Semi-structured interview guide used in nine qualitative interviews.

Paper	Study scenario	Data	Visualizations compared	Type	Sample size (valid)	How it was measured
Wainer & Francolini [1]	Census two-variable maps	R	Univariate and bivariate colour maps	PT	N: 16	Elementary map-reading error rate; response time
Sanyal et al. [2]	Abstract 1D and 2D datasets	S	Error bars, glyph size, glyph colour, surface colour	PT	N: 27	Search and counting accuracy; response time
MacEachren et al. [3]	Regional uncertainty maps	S	**E1,** Series 1: location, size, colour hue, colour value, colour saturation, orientation, grain, arrangement, shape, fuzziness, and transparency. **E2,** Series 1: fuzziness and colour value.	PT, SR	**E1:** 72; **E2:** 30	Intuitiveness ranking (from 1 to 7); comparison accuracy; response time
Correll & Gleicher [4]	Generic mean-and-interval judgments	S	Error bars, box plots, gradients, violin plots	PJ	**E1:** 96; **E2:** 48; **E3:** 96	Prediction accuracy; likelihood judgments; judgment confidence
Scholz & Lu [5]	Texas precipitation and crop siting	R	Boundary fuzziness, colour lightness, symbol shape, and symbol transparency	SR	N: 37	Subjective ranking
Greis et al. [6]	Weather forecasts in a farming game	M	Text: expected value, expected value + SD, and quantiles; graphics: line chart + CI, box-and-whisker plot, bar + error bars, histogram bars, stacked histogram, area histogram, shaded probability bars, PDF, and CDF	DM, SR	Survey: 90; game: 12	Subjective ratings; farming-game decision choices
Hullman et al. [7]	Fictitious sea-water-solute concentrations	S	HOPs, error bars, violin plots	PJ	**N:** 288	Probability-estimation error
Gschwandtner et al. [8]	Abstract uncertain time intervals	S	Gradient plots, violin plots, accumulated probability plots, error bars, centred error bars, ambiguation	PT, PJ, SR	N: 72	Start/end-time, duration, and probability-estimation error; completion time; five-point preference ratings
Correll et al. [9]	Abstract heatmaps and Battleship game	S	**E1:** VSUP + wedge legend; VSUP + square legend; Traditional bivariate + wedge legend; Traditional bivariate + square legend; Continuous bivariate + wedge legend; Continuous bivariate + square legend; Juxtaposed univariate + discrete colorbar; Juxtaposed univariate + continuous colorbar. **E2:** VSUP + wedge legend; VSUP + square legend; Traditional bivariate + wedge legend; Traditional bivariate + square legend.	PT, DM	**E1:** 24; **E2:** 21	Identification accuracy; response time; decision choices
Fernandes et al. [10]	Simulated real-time bus arrivals	S	Dotplots, CDFs, PDFs, intervals, text, baseline	DM	N: 408	Expected reward relative to optimal; variability across trials
Kale et al. [11]	Simulated monthly jobs-report trends	S	HOPs, error bars, line ensembles	PT	**E1:** 50; **E2:** 150	Trend-identification accuracy; perceptual sensitivity (JND); judgment confidence
Kusumastuti et al. [12]	Abstract bivariate Gaussian uncertainty	S	Ensemble, summary, distribution displays	DM	**E1:** 118; **E2:** 56	Decision accuracy; response time
Sterzik et al. [13]	Protein positional uncertainty	M	Colour, clouds, ensembles, hulls, sausages, contours, texture, waves, noise	PT, QF	**E1:** 540; **E2:** varies by metaphor (793 total); **E3:** 423; experts: 6	Mapping-direction judgments; uncertainty-estimation accuracy; qualitative expert feedback

Table S2: Empirical evaluations of uncertainty visualization techniques. This table and Table S3 summarize only the parts of each study that directly compare uncertainty visualisations; they do not describe the studies in full.
Data: S = synthetic or simulated data; R = real observational data; M = mixed real and simulated data;
Task type: PT = perceptual tasks; PJ = probability judgments; DM = decision-making tasks; SR = subjective ratings; QF = qualitative feedback.
N = number of participants. Sample size (valid) reports participants included in the relevant analysis rather than all participants recruited; separate values are shown when the analytic sample differed across experiments or conditions.
E1–E3 = separate experiments.

Paper	Experiment/task	Best-performing visualization	Other visualizations, in descending order
Wainer & Francolini [1]	**E1**†: Identify the level of both variables at a marked map location. **E2**†: Repeat the task without the colour legend.	**E1:** paired univariate maps; time: bivariate map. **E2:** paired univariate maps.	**E1**: bivariate map; time: paired univariate maps. **E2:** bivariate map.
Sanyal et al. [2]	**E1**†: Find the least or most uncertain value. **E2**†: Count values matching a specified data or uncertainty feature.	**E1:** *1D*: glyph size for locating the least uncertain value; glyph colour and surface colour for locating the most uncertain value. *2D*: surface colour for locating the least uncertain value; glyph colour and surface colour for locating the most uncertain value. **E2:** *1D*: no clear winner. *2D*: surface colour for counting uncertainty features; glyph size for counting data features.	**E1:** *1D* least: glyph colour ≈ surface colour > error bars; *1D* highest: glyph size > error bars; *2D* least: glyph size ≈ glyph colour ≈ error bars; *2D* highest: glyph size > error bars. **E2:** *1D*: all four performed similarly; *2D* uncertainty features: glyph size ≈ glyph colour ≈ error bars; *2D* data features: glyph colour ≈ error bars > surface colour.
MacEachren et al. [3]	**E1,** Series 1†: Rate the intuitiveness of 22 abstract symbol sets (11 visual variables × 2 encoding directions) for representing three ordered levels of uncertainty, using a 1–7 scale. **E2,** Series 1†: Compare two map regions, each containing nine locations with three possible uncertainty levels, and select the region with greater overall uncertainty.	**E1,** Series 1: fuzziness, location, and value; response time: no clear winner. **E2,** Series 1: no clear winner for accuracy or response time.	**E1,** Series 1: arrangement, size, and transparency > saturation, hue, orientation, and shape. **E2,** Series 1: fuzziness ≈ colour value.
Correll & Gleicher [4]	**E1**†: A chart showed an estimate and a marked possible outcome. Participants predicted whether the actual outcome would be higher or lower, rated their confidence, and judged how likely the marked outcome was. **E2**†: Judge a proposed outcome when the outcome and/or margin of error was presented as text. **E3**†: Compare two uncertain groups, predict which will have the higher outcome, rate confidence, and estimate how large the difference will be.	**E1:** gradient ≈ violin ≈ box plot. **E2:** no clear winner. **E3:** gradient ≈ violin ≈ box plot.	**E1:** bar chart with error bars; values inside the bar were judged more likely. **E2:** moving both elements to text reduced within-bar bias but also reduced accuracy and increased unjustified confidence. **E3:** bar chart with error bars; participants predicted larger group differences and reported greater confidence.
Scholz & Lu [5]	**E1**†: Rank four uncertainty visualisations from most to least preferred.	**E1:** boundary fuzziness (mean rank 3.35).	**E1:** colour lightness (2.97) > symbol transparency (2.38) > symbol shape (1.30).
Greis et al. [6]	**E1**†: Rate how well 12 weather-forecast displays communicated uncertainty. **E2:** Use a weather forecast to choose a crop in a farming game, then select a preferred display.	**E1:** line chart with confidence interval (4.00/5). **E2:** probability-density function (69% optimal; preferred by 8/12).	**E1:** histogram bars (3.87) > PDF (3.71) > expected value (3.63) > stacked histogram (3.41) > CDF (3.34) > shaded bars (3.31) > mean + SD (3.26) > bar + error bars (3.16) > area histogram (3.04) > boxplot (2.74) > quantiles (2.43). **E2** decisions: histogram bars (64%) > expected value (60%) > line + CI (57%).
Hullman et al. [7]	**E1**†: Estimate properties of one uncertain quantity. **E2**†: Estimate how often B > A for two uncertain quantities. **E3**†: Estimate how often B is largest for three uncertain quantities.	**E1:** no clear winner. **E2–E3:** HOPs.	**E2–E3:** violin ≈ error bars.
Gschwandtner et al. [8]	**E2**†: Effectiveness in representing start and end time. **E3a**†: Effectiveness in representing minimum and maximum duration. **E3b**†: Effectiveness in representing average duration. **E4**†: Effectiveness in representing probability of start and end time. **E5:** Preferences about visualization types.	**E2:** ambiguation ≈ error bars ≈ centred error bars. **E3a:** ambiguation ≈ error bars ≈ centred error bars ≈ gradient. **E3b:** centred error bars ≈ error bars ≈ ambiguation. **E4:** gradient. **E5:** ambiguation.	**E2:** gradient > accumulated probability ≈ violin. **E3a:** accumulated probability ≈ violin. **E3b:** gradient, accumulated probability, and violin. **E4:** accumulated probability ≈ violin. **E5:** accumulated probability > centred error bars ≈ error bars > violin > gradient.
Correll et al. [9]	**E1**†: Find the grid cell matching a specified value and uncertainty. **E2**†: Place a ship where value was high while avoiding uncertain locations.	**E1:** VSUP + square legend ≈ VSUP + wedge legend ≈ Traditional bivariate + wedge legend ≈ Traditional bivariate + square legend. **E2:** VSUP + wedge legend ≈ VSUP + square legend.	**E1:** Juxtaposed univariate + discrete colorbar > Juxtaposed univariate + continuous colorbar > Continuous bivariate + wedge legend > Continuous bivariate + square legend. **E2:** Traditional bivariate + wedge legend ≈ Traditional bivariate + square legend.
Fernandes et al. [10]	**E1:** Choose when to leave to catch a bus while balancing waiting cost against the risk of missing it.	**E1:** 50-dot quantile dotplot.	**E1:** CDF ≈ 20-dot quantile dotplot > remaining displays; the complete lower-order ranking was not reported, and high-probability text approached their mean performance with greater variability.
Kale et al. [11]	**E1**†: Decide whether noisy employment reports showed growth or no growth. **E2**†: Repeat the trend decision with different displays and report confidence.	**E1:** HOPs. **E2:** regular HOPs; no clear winner for confidence calibration.	**E1:** error bars. **E2:** fast HOPs ≈ line ensemble.
Kusumastuti et al. [12]	**E1** and **E2:** Combine two or three uncertain location estimates to select the most likely hidden location, before and after practice.	**E1** two-source accuracy: ellipse ≈ interlace ≈ scatter; three-source: scatter ≈ ellipse. **E2** three-source accuracy: ellipse; time in both experiments: interlace ≈ scatter.	**E1** three-source accuracy: interlace. **E2** three-source accuracy: interlace ≈ scatter; time: ellipse.
Sterzik et al. [13]	**E1:** State what each metaphor appeared to represent without instruction. **E2:** Identify the region that appeared most uncertain. **E3:** Estimate the uncertainty value at a highlighted position.	**E1:** no clear winner. **E2:** noise (approximately 90%). **E3:** texture (lowest error; most well-rounded).	**E2** approximate partial order: texture > colour ≈ waves ≈ contours > hulls > ensemble ≈ clouds > sausages. **E3:** the remaining order changed by molecular representation; noise, waves, and colour generally formed the next group.

Table S3: Task-specific outcomes from the evaluations summarized in Table S2. **E** labels distinguish task groups, experiments, or studies within a paper. Rankings follow reported point estimates and authors' stated comparisons; they do not imply statistically significant differences. ">" means better performance for the stated task, "≈" denotes a reported tie, an inconclusive contrast, or no reliable difference, and "not reported" indicates that the source did not support a complete ranking.

† An inferential significance test of visualization differences was reported. The symbol does not imply that every displayed ordering was significant; unmarked evaluations may still report estimates and confidence or credible intervals.

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